

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4 Total Marks:40

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.

2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.
5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATI ON Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understand ing of Concepts	25	Demonstrat es thorough understandi ng and effectively integrates multiple concepts/te chnologies	Good understandi ng with proper integration of most concepts	Basic understandi ng; limited or partial integration	Poor understandi ng; unable to integrate concepts
Experiment al Design	30	Well- planned, systematic implementa tion with optimal solution	Correct implementa tion with minor issues	Basic implementa tion with errors or inefficiencie s	Incorrect or incomplete implementa tion

Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

QUESTION 2

Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.

AIM:

To create a network topology consisting of two LANs connected through a router using Cisco Packet Tracer, configure IPv4 addresses, subnet masks, and default gateways, and verify end-to-end communication using the ping command. To capture ICMP packets using Wireshark, analyze the IPv4 header fields, and understand the role of the Network layer in forwarding packets between different networks.

SOFTWARE / REQUIREMENTS:

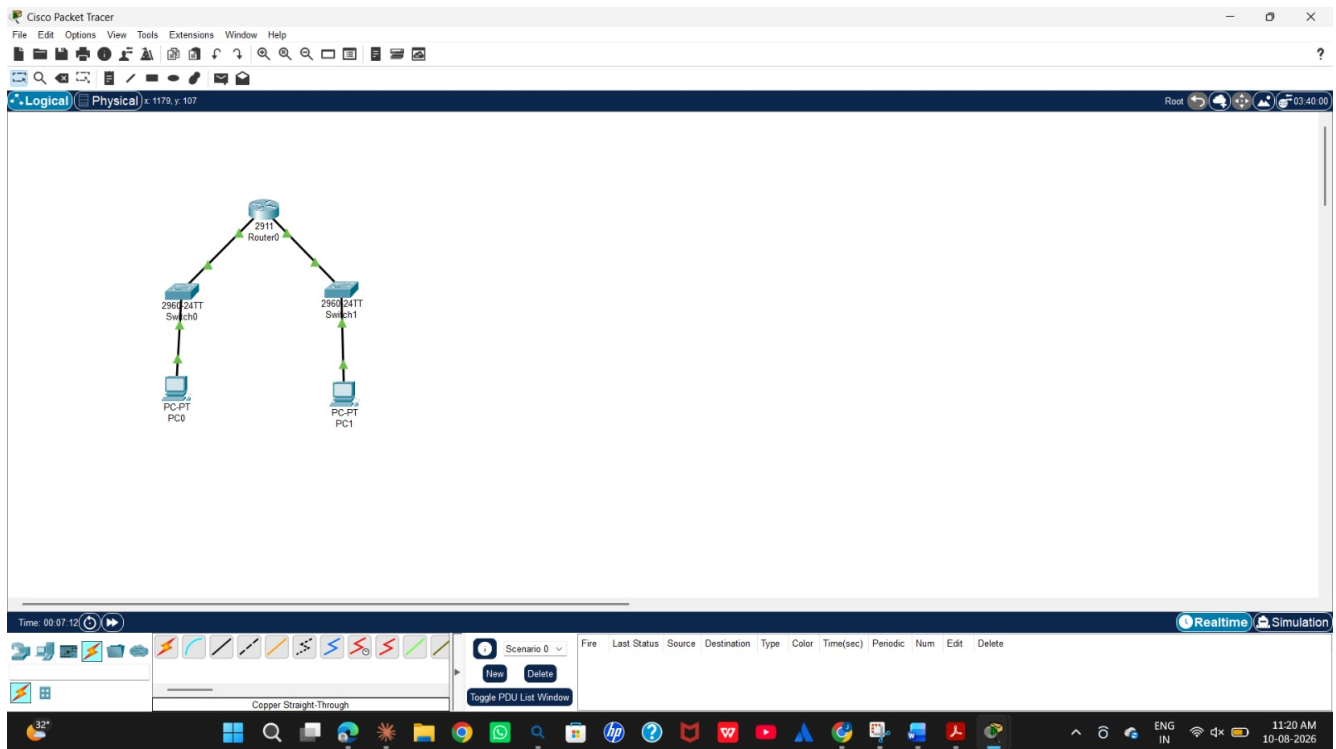
Cisco Packet Tracer, Wireshark, Two PCs, One Cisco Router, Ethernet cables.

PROCEDURE:

1. Open Cisco Packet Tracer and create a new workspace.
2. Add two PCs and one router to the workspace.
3. Connect PC0 to the first router interface and PC1 to the second router interface using Ethernet cables.
4. Configure the router interface connected to PC0 with IP address 192.168.1.1 and subnet mask 255.255.255.0.
5. Configure the router interface connected to PC1 with IP address 192.168.2.1 and subnet mask 255.255.255.0.
6. Enable both router interfaces using the no shutdown command.
7. Configure PC0 with IP address 192.168.1.10, subnet mask 255.255.255.0, and default gateway 192.168.1.1.
8. Configure PC1 with IP address 192.168.2.10, subnet mask 255.255.255.0, and default gateway 192.168.2.1.
9. Verify the router configuration using the show ip route command.
10. From PC0, use the ping command to test communication with PC1.
11. Verify that successful Echo Reply messages are received with zero packet loss.
12. Open Wireshark and select the active Wi-Fi interface.
13. Start the Wireshark capture and apply the display filter icmp.
14. Generate ICMP traffic and capture the packets.
15. Select an IPv4 ICMP packet and analyze the Source IP Address, Destination IP Address, TTL, Protocol Number, and Header Length.
16. Explain how the Network layer forwards packets between the two different networks.

Network Topology:

The topology consists of two LANs connected through a central router. PC0 belongs to the 192.168.1.0/24 network and PC1 belongs to the 192.168.2.0/24 network. The router connects the two different networks and forwards packets between them.



Router Configuration:

The router interfaces were configured with IP addresses belonging to two different networks.

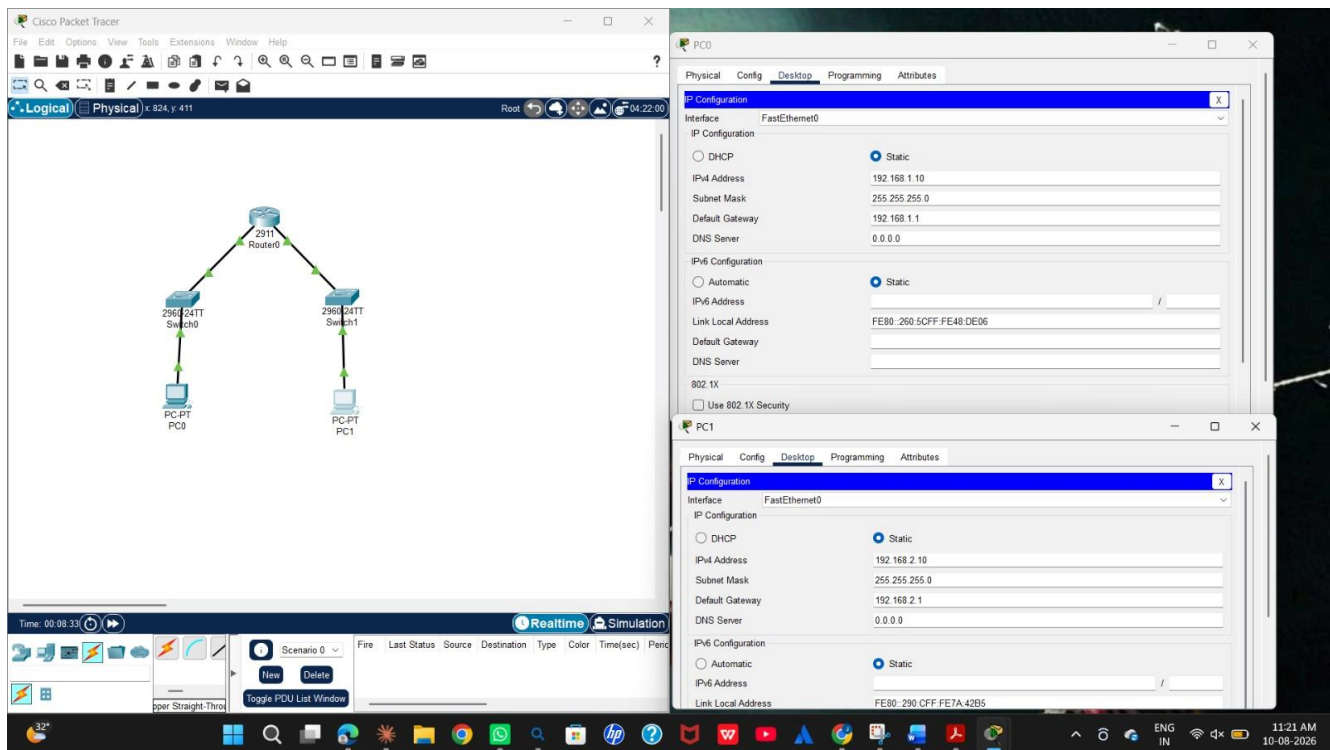
Router G0/0:

IP	Address:	192.168.1.1
Subnet Mask: 255.255.255.0		

Router G0/1:

IP	Address:	192.168.2.1
Subnet Mask: 255.255.255.0		

The interfaces were enabled using the no shutdown command. The show ip route command was used to verify that both networks were directly connected to the router.



PC0 Configuration:

PC0 was assigned an IP address in the first network.

IP	Address:	192.168.1.10
Subnet	Mask:	255.255.255.0
Default Gateway: 192.168.1.1		

The default gateway is the router interface connected to the first LAN. It allows PC0 to communicate with devices outside its local network.

PC1 Configuration:

PC1 was assigned an IP address in the second network.

IP	Address:	192.168.2.10
Subnet	Mask:	255.255.255.0
Default Gateway: 192.168.2.1		

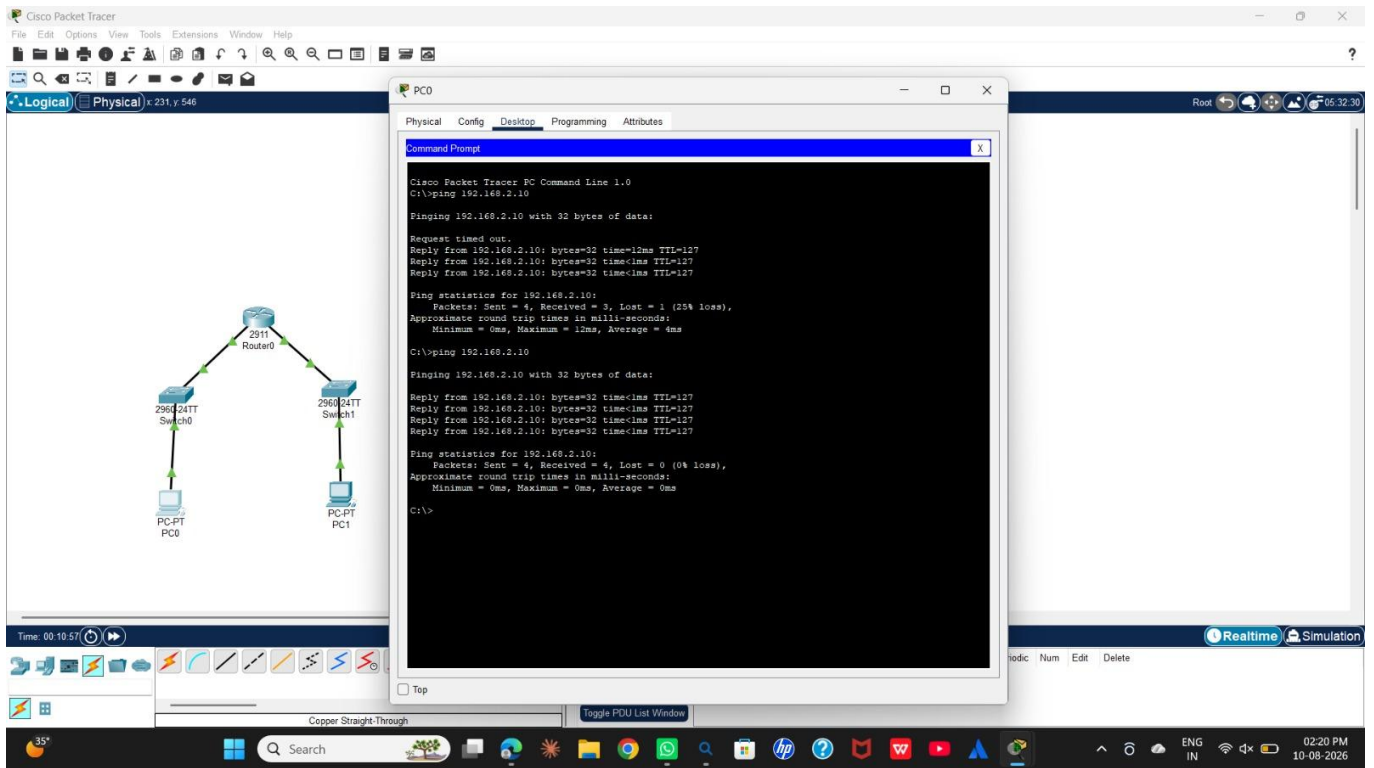
The default gateway is the router interface connected to the second LAN.

PC0 to PC1 Connectivity:

The ping command was used from PC0 to verify communication with PC1.

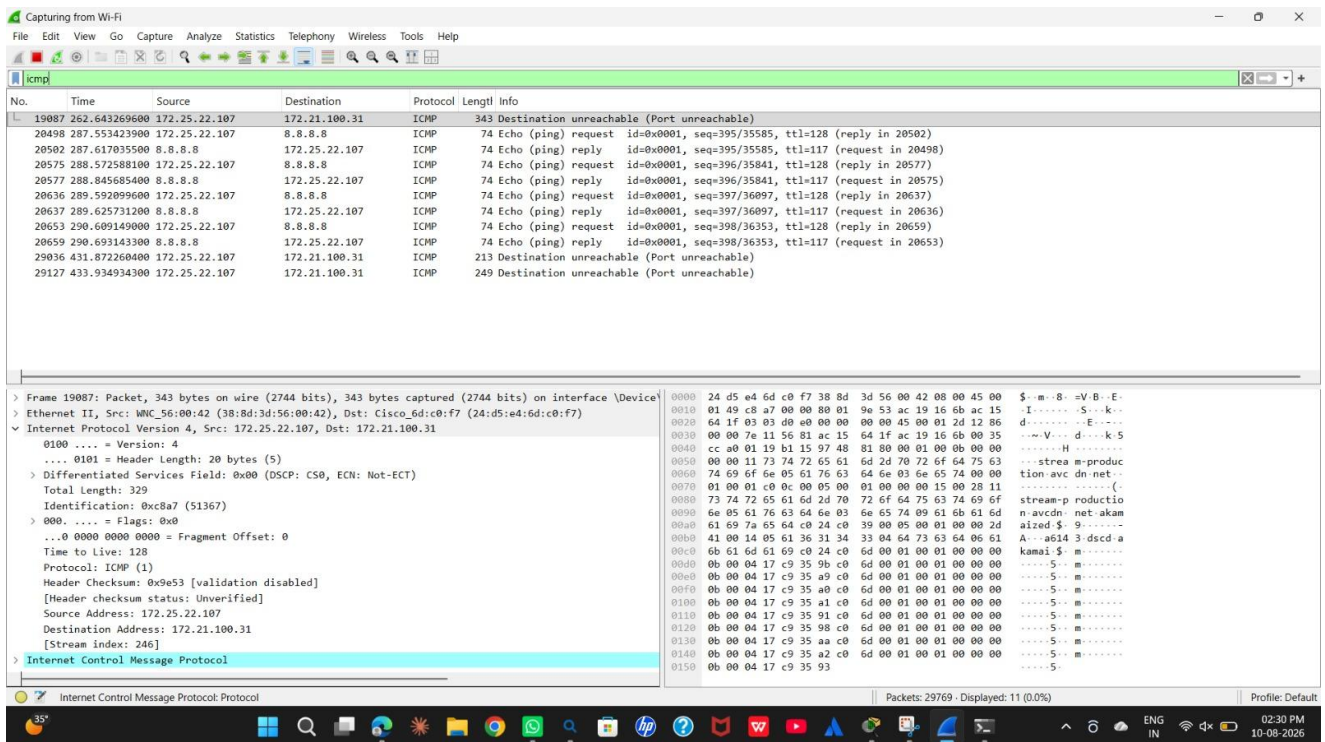
```
ping 192.168.2.10
```

Successful Echo Reply messages were received from PC1 and the packet loss was 0%. This confirms that the two different networks can communicate through the router.



WIRESHARK OUTPUT:

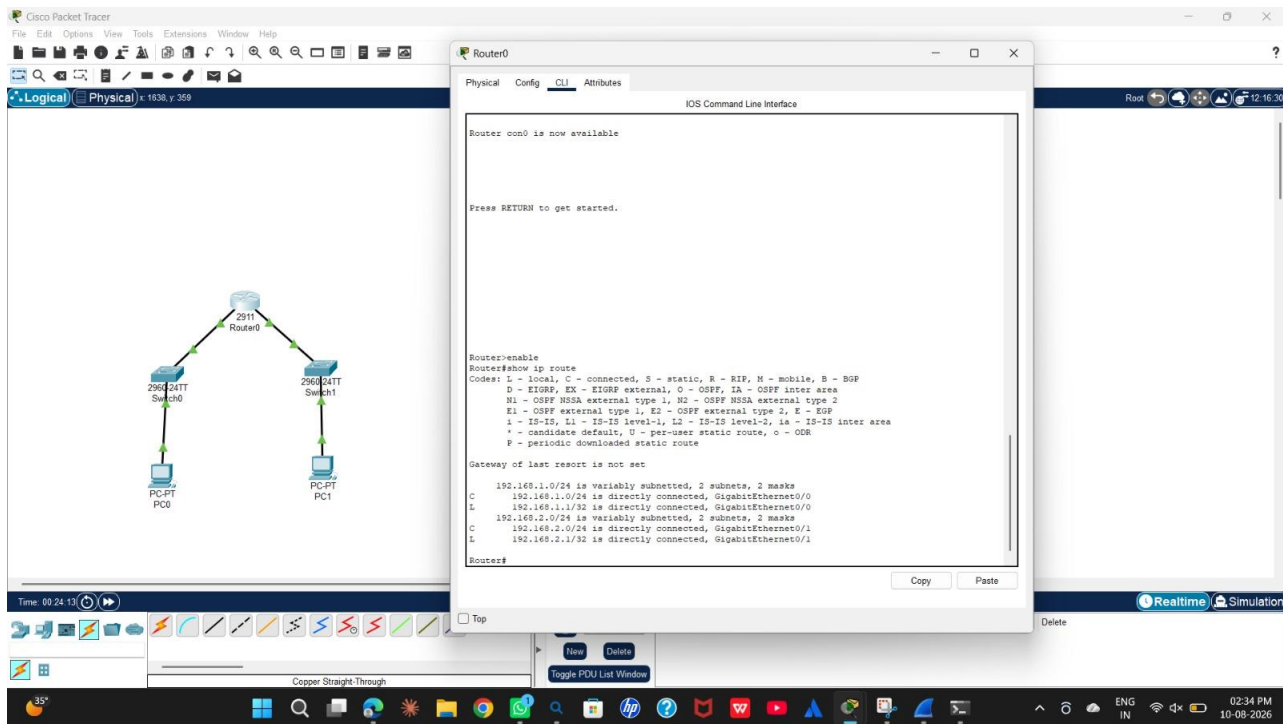
Wireshark was used to capture ICMP traffic. The icmp display filter was applied to display only ICMP packets. The captured packets were examined to identify the IPv4 header information associated with the ICMP communication.



IPv4 Header Analysis:

IPv4 Header Field	Description
Source IP Address	IP address of the device sending the packet.
Destination IP Address	IP address of the device receiving the packet.
TTL	Limits the lifetime of the packet.
Protocol	Identifies the upper-layer protocol; ICMP = 1.
Header Length	Size of the IPv4 header; normally 20 bytes when no options are present.

The Source IP Address identifies the sender of the packet, while the Destination IP Address identifies the intended receiver. The Time-To-Live (TTL) value prevents packets from circulating indefinitely in the network. Each router that forwards the packet decreases the TTL value. The Protocol Number in an ICMP packet is 1, indicating that the payload contains ICMP. The normal IPv4 Header Length is 20 bytes, represented by an Internet Header Length (IHL) value of 5.



Role of the Network Layer:

The Network layer is responsible for logical addressing and routing packets between different networks. IPv4 provides logical IP addresses that identify the source and destination devices. When PC0 sends a packet to PC1, the destination IP address belongs to a different network. Therefore, PC0 sends the packet to its default gateway, 192.168.1.1. The router examines the destination IP address and its routing table, determines that the 192.168.2.0/24 network is connected through its second interface, and forwards the packet through that interface. Thus, the Network layer enables communication between different networks by using IP addressing, routing, and packet forwarding.

RESULT:

Thus, two LANs connected through a router were successfully created and configured in Cisco Packet Tracer. IPv4 addresses, subnet masks, and default gateways were configured for all devices. End-to-end communication was successfully verified using the ping command with zero packet loss. ICMP packets were captured using Wireshark and the important IPv4 header fields including Source IP Address, Destination IP Address, TTL, Protocol Number, and Header Length were analyzed successfully. The role of the Network layer in forwarding packets between different networks was also understood.

QUESTION 4

Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

AIM:

To develop a simple network consisting of two PCs connected through a router using Cisco Packet Tracer, configure the required IP addresses, and verify connectivity using the ping command. To capture and analyze ICMP Echo Request and Echo Reply messages using Wireshark, examine important ICMP fields, and calculate the Round Trip Time (RTT).

SOFTWARE / REQUIREMENTS:

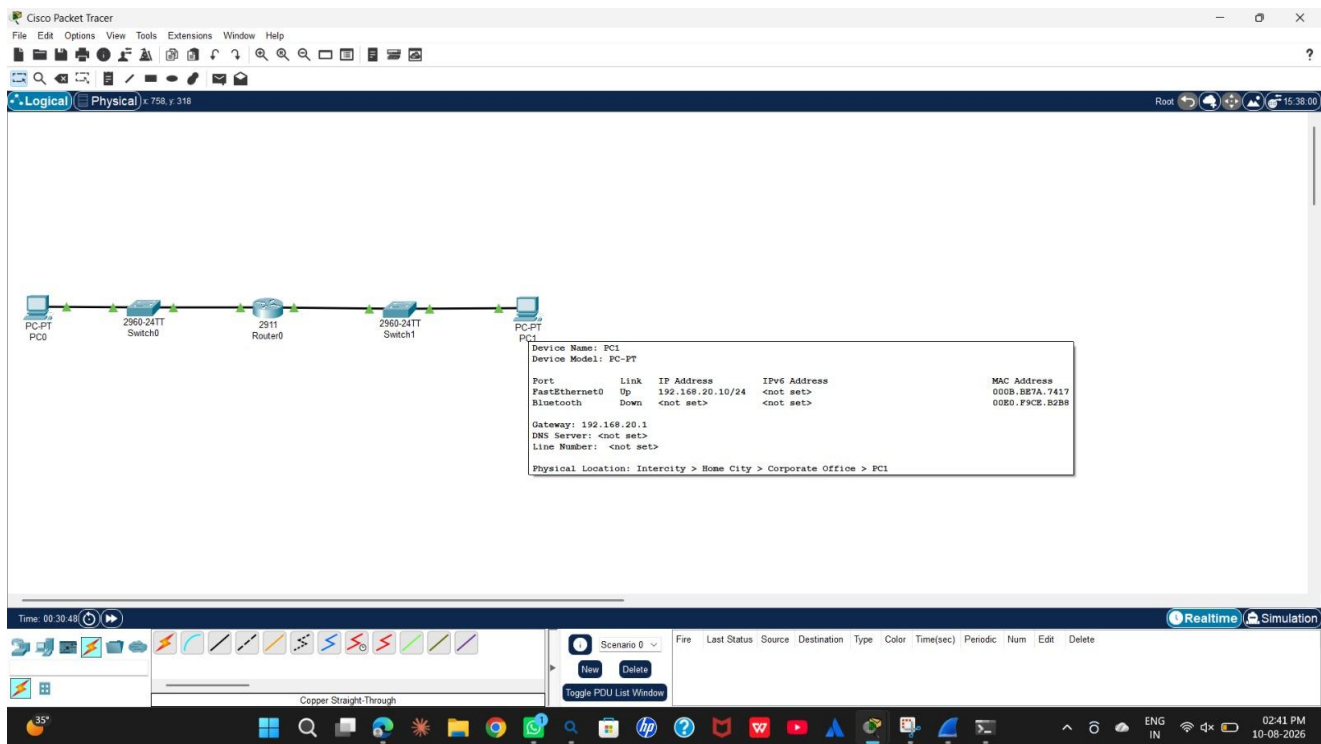
Cisco Packet Tracer, Wireshark, Two PCs, One Cisco Router, Ethernet cables.

PROCEDURE:

17. Open Cisco Packet Tracer and create a new workspace.
18. Add two PCs and one router.
19. Connect PC0 to the first router interface and PC1 to the second router interface using Ethernet cables.
20. Configure the router interfaces with suitable IP addresses belonging to two different networks.
21. Enable the router interfaces using the no shutdown command.
22. Configure the IP address, subnet mask, and default gateway of PC0.
23. Configure the IP address, subnet mask, and default gateway of PC1.
24. Verify the router interface configuration.
25. From PC0, use the ping command to test connectivity with PC1.
26. Verify that successful Echo Reply messages are received with zero packet loss.
27. Open Wireshark and select the active Wi-Fi interface.
28. Start the Wireshark capture.
29. Apply the display filter icmp.
30. Generate ICMP traffic using the ping command.
31. Identify the ICMP Echo Request and Echo Reply packets.
32. Examine the Type, Code, Identifier, and Sequence Number fields.
33. Select a matching Echo Request and Echo Reply pair.
34. Calculate the Round Trip Time (RTT) using the request and reply timestamps.

Network Topology:

The topology consists of two PCs connected to a central router. The two PCs are configured in different IP networks, and the router provides communication between the two networks.



Router Configuration:

The router interfaces were configured with IP addresses belonging to two different networks.

For the first network, the router interface was configured as:

IP	Address:	192.168.10.1
Subnet Mask: 255.255.255.0		

For the second network, the router interface was configured as:

IP	Address:	192.168.20.1
Subnet Mask: 255.255.255.0		

Both interfaces were enabled using the no shutdown command.

PC0 Configuration:

IP	Address:	192.168.10.10
Subnet	Mask:	255.255.255.0
Default Gateway: 192.168.10.1		

PC1 Configuration:

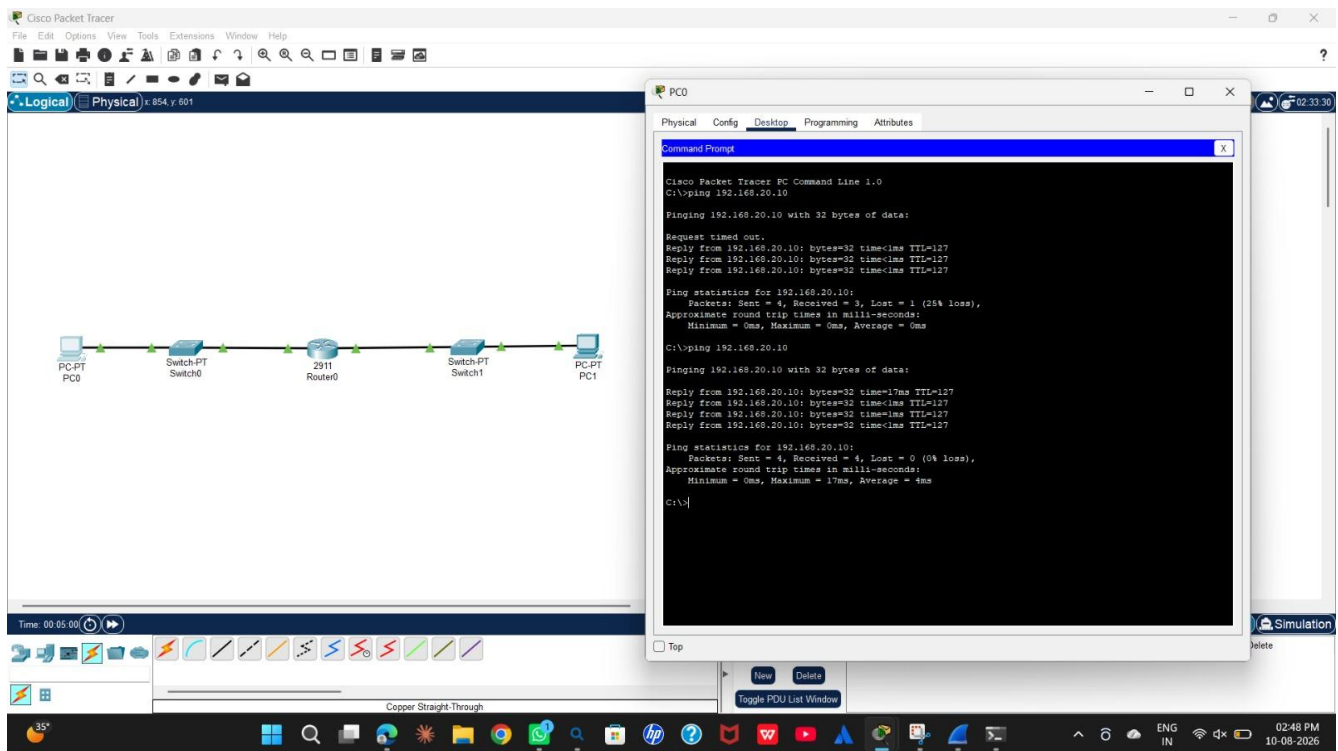
IP	Address:	192.168.20.10
Subnet	Mask:	255.255.255.0
Default Gateway: 192.168.20.1		

PC0 to PC1 Connectivity:

The ping command was used from PC0 to test communication with PC1.

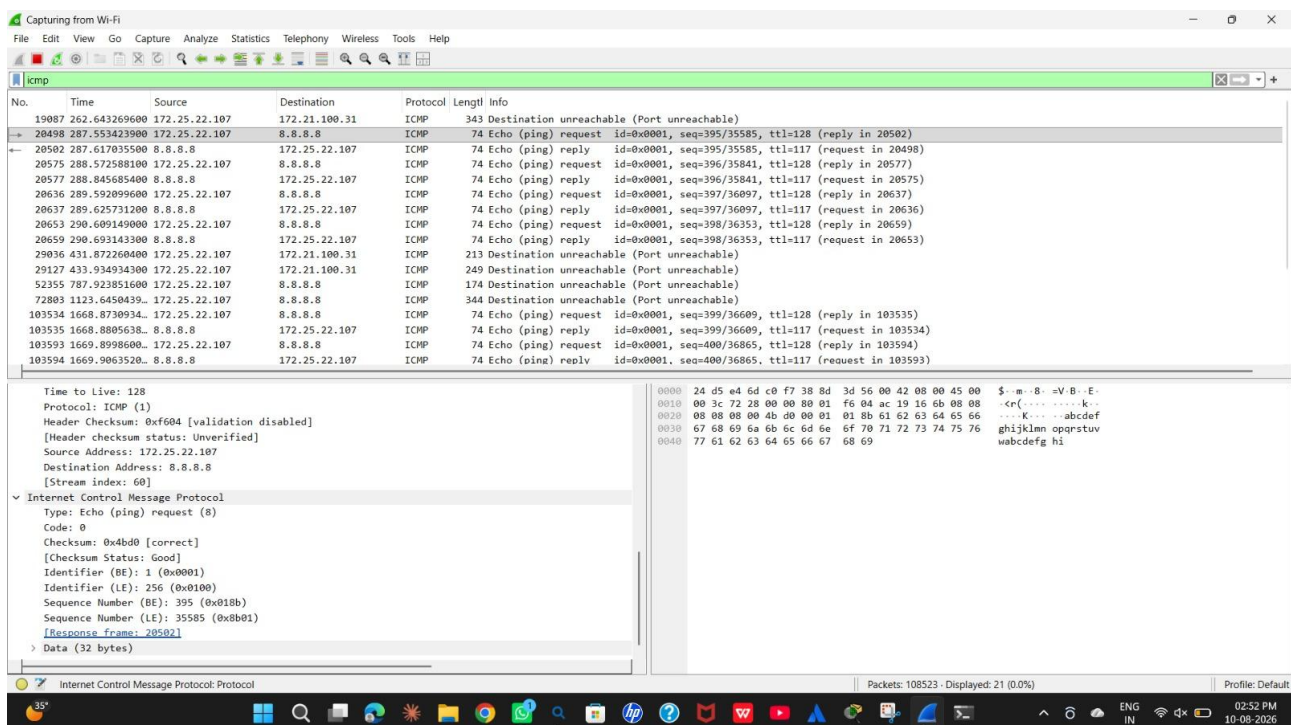
```
ping 192.168.20.10
```

The ping test produced successful Echo Reply messages and showed 0% packet loss. This confirms that PC0 and PC1 can communicate through the router.



WIRESHARK OUTPUT:

Wireshark was used to capture the ICMP traffic. The display filter icmp was applied to display only ICMP packets. The captured traffic contained ICMP Echo Request and Echo Reply packets generated by the ping operation.



ICMP Echo Request Analysis:

The selected packet is an ICMP Echo Request. The important ICMP fields are:

Field	Value
Type	8
Code	0
Identifier	0x0001
Sequence Number	395/35585

Type 8 represents an ICMP Echo Request. Code 0 indicates that there is no additional error code. The Identifier identifies the ping process, while the Sequence Number identifies an individual Echo Request and helps associate it with the corresponding Echo Reply.

ICMP Echo Reply Analysis:

The selected matching packet is an ICMP Echo Reply. The important fields are:

Field	Value
Type	0
Code	0
Identifier	0x0001
Sequence Number	395/35585

Type 0 represents an ICMP Echo Reply. The Identifier and Sequence Number match those of the corresponding Echo Request, confirming that the packets belong to the same ping exchange. In the Wireshark capture, the response time for the selected request-reply pair was 63.612 ms.

The image shows a Wireshark packet capture window titled "Capturing from Wi-Fi". The packet list on the left shows a series of ICMP Echo (ping) requests and replies. The selected packet is an ICMP Echo Reply (Type 0) with ID 20498, source 8.8.8.8, and destination 172.25.22.107. The packet details pane on the right shows the ICMP header and the payload, which is the response to the request. The response time is 63.612 ms.

Time to Live: 117
Protocol: ICMP (1)
Header Checksum: 0x72cd [validation disabled]
[Header checksum status: Unverified]
Source Address: 8.8.8.8
Destination Address: 172.25.22.107
[Stream index: 60]
Internet Control Message Protocol
Type: Echo (ping) reply (0)
Code: 0
Checksum: 0x53d0 [correct]
[Checksum Status: Good]
Identifier (BE): 1 (0x0001)
Identifier (LE): 256 (0x0100)
Sequence Number (BE): 395 (0x018b)
Sequence Number (LE): 35585 (0x8bb1)
[Request frame: 20498]
[Response time: 63.612 ms]

0000 38 8d 3d 56 00 42 24 d5 e4 6d c0 f7 08 00 45 60 8 ->V.B5- -m---E-
0010 00 3c 00 00 00 00 75 01 72 cd 08 08 08 08 ac 19 -<---u- r-----
0020 16 6b 00 00 53 00 00 01 8b 61 62 63 64 65 66 -<---S- -akcdef
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklm opqrstuv
0040 77 61 62 63 64 65 66 67 68 69 wabcdefg hi

Calculation:

For the selected Echo Request and Echo Reply pair:

$$RTT = TReply - TRequest$$

From the Wireshark capture, the response time was:

$$RTT = 63.612 \text{ ms}$$

Therefore, $RTT = 63.612 \text{ ms}$. The RTT represents the total time taken for the Echo Request to travel from the source to the destination and for the Echo Reply to return to the source.

Echo Request and Echo Reply:

The matching ICMP packets were identified using the same Identifier and Sequence Number.

The Echo Request contains:

Type	=	8
Code	=	0
Identifier	=	0x0001
Sequence Number	=	395/35585

The corresponding Echo Reply contains:

Type	=	0
Code	=	0
Identifier	=	0x0001
Sequence Number	=	395/35585

The matching Identifier and Sequence Number confirm that the request and reply belong to the same communication exchange.

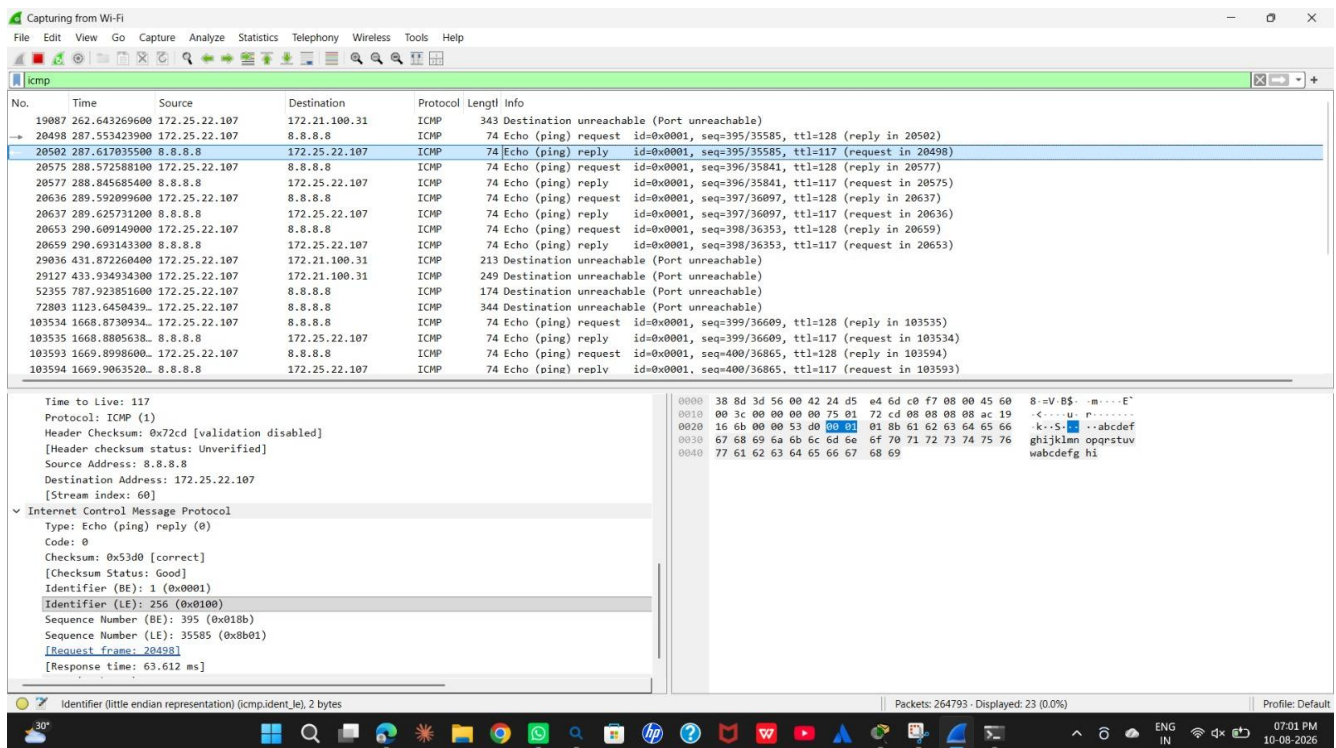
Wireshark packet capture showing ICMP Echo (ping) request and reply. The packet list shows a request from 172.25.22.107 to 8.8.8.8 and a reply from 8.8.8.8 to 172.25.22.107. The packet details pane shows the ICMP Echo (ping) request with Identifier (BE): 1 (0x0001) and Sequence Number (BE): 395 (0x018b). The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
19087	262.643269600	172.25.22.107	172.21.100.31	ICMP	343	Destination unreachable (Port unreachable)
20498	287.553423900	172.25.22.107	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=395/35585, ttl=128 (reply in 20502)
20502	287.617035500	8.8.8.8	172.25.22.107	ICMP	74	Echo (ping) reply id=0x0001, seq=395/35585, ttl=117 (request in 20498)
20575	288.572580100	172.25.22.107	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=396/35841, ttl=128 (reply in 20577)
20577	288.845685400	8.8.8.8	172.25.22.107	ICMP	74	Echo (ping) reply id=0x0001, seq=396/35841, ttl=117 (request in 20575)
20636	289.592099600	172.25.22.107	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=397/36097, ttl=128 (reply in 20637)
20637	289.625731200	8.8.8.8	172.25.22.107	ICMP	74	Echo (ping) reply id=0x0001, seq=397/36097, ttl=117 (request in 20636)
20653	290.609149000	172.25.22.107	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=398/36353, ttl=128 (reply in 20659)
20659	290.609143300	8.8.8.8	172.25.22.107	ICMP	74	Echo (ping) reply id=0x0001, seq=398/36353, ttl=117 (request in 20653)
29036	431.872260400	172.25.22.107	172.21.100.31	ICMP	213	Destination unreachable (Port unreachable)
29127	433.934934300	172.25.22.107	172.21.100.31	ICMP	249	Destination unreachable (Port unreachable)
52355	787.923851600	172.25.22.107	8.8.8.8	ICMP	174	Destination unreachable (Port unreachable)
72803	1123.6450439..	172.25.22.107	8.8.8.8	ICMP	344	Destination unreachable (Port unreachable)
103534	1668.8730934..	172.25.22.107	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=399/36609, ttl=128 (reply in 103535)
103535	1668.8805638..	8.8.8.8	172.25.22.107	ICMP	74	Echo (ping) reply id=0x0001, seq=399/36609, ttl=117 (request in 103534)
103593	1669.8998600..	172.25.22.107	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=400/36865, ttl=128 (reply in 103594)
103594	1669.9063520..	8.8.8.8	172.25.22.107	ICMP	74	Echo (ping) reply id=0x0001, seq=400/36865, ttl=117 (request in 103593)

Time to Live: 128
 Protocol: ICMP (1)
 Header Checksum: 0xf604 [validation disabled]
 [Header checksum status: Unverified]
 Source Address: 172.25.22.107
 Destination Address: 8.8.8.8
 [Stream index: 60]

Internet Control Message Protocol
 Type: Echo (ping) request (8)
 Code: 0
 Checksum: 0x4bd0 [correct]
 [Checksum Status: Good]
 Identifier (BE): 1 (0x0001)
 Identifier (LE): 256 (0x0100)
 Sequence Number (BE): 395 (0x018b)
 Sequence Number (LE): 35585 (0x0801)
 [Response frame: 20502]
 Data (32 bytes)

0000 24 d5 e4 6d c0 f7 38 8d 3d 56 00 42 08 00 45 00 \$-m-B-v-B-E-
 0010 00 3c 72 28 00 00 80 01 f6 04 ac 19 16 6b 08 08 <-(---k-
 0020 08 08 08 00 4b d0 00 01 01 8b 61 62 63 64 65 66 ...K-K-..abcdef
 0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv
 0040 77 61 62 63 64 65 66 67 68 69 wabcdefg hi



Role of ICMP in Network Troubleshooting:

ICMP (Internet Control Message Protocol) is used for network connectivity verification and troubleshooting. The ping command uses ICMP Echo Request and Echo Reply messages to determine whether a destination is reachable.

- Verify connectivity between two devices.
- Identify packet loss.
- Measure network latency using RTT.
- Detect unreachable destinations.
- Diagnose network and routing problems.
- Verify whether a device is responding to network requests.

When an Echo Request is sent and an Echo Reply is received, it confirms that communication is possible between the source and destination. The RTT provides information about the delay in the network.

RESULT:

Thus, a simple network consisting of two PCs connected through a router was successfully created and configured in Cisco Packet Tracer. Connectivity was successfully verified using the ping command with zero packet loss. ICMP Echo Request and Echo Reply packets were captured and analyzed using Wireshark. The Type, Code, Identifier, and Sequence Number fields were successfully identified. The Round Trip Time for the selected request-reply pair was calculated as 63.612 ms. ICMP was also studied as an important protocol for network troubleshooting and connectivity verification.